Lab: Expected Free Energy and Policy Selection

> **Learning Goal:** Compute and decompose EFE for simple policies and analyze exploration-exploitation trade-offs.

Part 1: EFE for Simple Policies

Scenario: A foraging agent with two policies:

- π_1: Go to known food source (certain food, no new information)
- π_2: Explore unknown area (uncertain food, high information gain)

Preferences: C = [food: +3, no_food: -1]

1. For π_1: Expected observation = food (100% certain). Compute pragmatic value.
2. For π_1: Information gain = 0 (already knows this area). Compute epistemic value.
3. For π_2: Expected observation = food (50%) or no_food (50%). Compute pragmatic value.
4. For π_2: Information gain = high (will learn about new area). Estimate epistemic value as +2.
5. Compute $G(\pi)$ for each policy. Which policy is selected?

Write your response here

Part 2: Exploration-Exploitation Dynamics

> **Learning Goal:** See how the balance shifts with uncertainty.

Exercise: The same agent, but now with varying uncertainty about the known food source:

Certainty about known source	Pragmatic value (π_1)	Epistemic value (π_2)	Which policy wins?
100% certain food there	+3	+2	
70% certain food there	+1.8	+2	
50% certain food there	+1	+2	

What pattern do you see? How does uncertainty shift the balance?

Write your response here

Part 3: Policy Precision (γ)

> **Learning Goal:** Understand how γ affects action selection.

Exercise: Two policies with $G(\pi_1) = -5$ and $G(\pi_2) = -3$ (lower is better, so π_1 is preferred).

Compute $P(\pi_1) = \text{softmax}(-\gamma \times G(\pi))$ for different γ values:

γ	$P(\pi_1)$	$P(\pi_2)$	How decisive?
0.1			
1.0			
10.0			

Hint: $\text{softmax}(-\gamma \times G) = \exp(-\gamma G) / \Sigma\exp(-\gamma G)$

Write your response here

Part 4: Flat C Vector — Pure Curiosity

Learning Goal: Analyze what happens when there are no preferences.

Exercise: If C = [0, 0, 0] (no preferences), what is the pragmatic value for any policy?

- If pragmatic value is zero, what is the only remaining drive?
- What kind of behavior would this agent exhibit?
- Is this a good model of infant exploration?

Write your response here

Part 5: Reflection

In 150 words, reflect: The EFE decomposition shows that a single mathematical quantity (Expected Free Energy) can explain both goal-directed behavior and curiosity. Why is this unification theoretically important? Are there behaviors this framework cannot explain?

Write your response here

Lab Summary

Part	Skill Practiced	Key Concept
1	EFE computation	Pragmatic + Epistemic decomposition
2	Trade-off analysis	Exploration-exploitation dynamics
3	Softmax computation	Policy precision γ
4	Edge case analysis	Pure curiosity agents
5	Theoretical reflection	Unification of goals and curiosity